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Machine-learning-based cost prediction models for inpatients with mental disorders in China

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Abstract

Background Mental disorders are increasingly prevalent, leading to increased medical expenditures. To refine the reimbursement of medical costs for inpatients with mental disorders by health insurance, an accurate prediction model is essential. Per-diem payment is a common internationally implemented payment method for medical insurance of inpatients with mental disorders, necessitating the exploration of advanced machine learning methods for predicting the average daily hospitalization costs (ADHC) based on the characteristics of inpatients with mental disorders.

Methods We used data including demographic information, clinical/functional characteristics, institutional features, and cost information of 5070 hospitalized patients with mental disorders in Jinhua, China, and employed six algorithms to predict ADHC. Performance of these six algorithms was evaluated through 5- old cross-validation combined with bootstrap method to select the most suitable algorithm and identify key factors influencing ADHC.

Results The random forest (RF) model demonstrated better performance (R-squared (R^2) = 0.6417 (95% CI, 0.6236–0.6611), root-mean-square error (RMSE) = 0.2398 (95% CI, 0.2252–0.2553), mean-absolute error (MAE) = 0.1677 (95% CI, 0.1626–0.1735), mean-absolute-percentage error (MAPE) = 0.0295 (95% CI, 0.0287–0.0304)). According to feature importance ranking, models incorporating top 11 factors (> 0.01) demonstrated comparable performance to those encompassing all variables. Top four factors (> 0.05) were level of medical institution, age, functional classification, and cognitive classification. Notably, level of medical institutions was the most significant factor across all primary models. Higher medical institutions level, patients below 20 and above 75 years old, lower functional classification, and lower cognitive classification are associated with increased ADHC.

Conclusions Machine learning algorithms, particularly RF algorithm, enhance accuracy of predicting ADHC for mental health patients. The findings of this study provide evidence for setting up more reasonable insurance payment standards for inpatients with mental disorders and support resource allocation in clinical practice.

Keywords Machine learning, Inpatients with mental disorders, Cost prediction

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Background

The prevalence of mental disorders is ubiquitous globally, with estimated one in three individuals experiencing a mental disorder in their lifetime [1–3]. Global prevalence of mental disorders increased by 48.1% from 1990 to 2019 [4]. In 2019, approximately 970 million people worldwide suffered from mental disorders, with anxiety and depression being the most common conditions [5]. Additionally, approximately 20% of children and adolescents exhibited mental health issues [6]. From 1990 to 2019, 8.8% of children and adolescents globally were diagnosed with various mental disorders [7]. Mental and substance use disorders accounted for 14.4% of persons with disabilities globally in 2017, ranking second among all categories [8]. In China, the lifetime prevalence of mental disorders among adults reached 16.6% in 2022, while the prevalence among children and adolescents aged 6–16 years was 17.5% in 2020 [9].

Mental illness has a considerable economic cost to individuals and society. The loss of productivity caused by mental illness greatly exceeds the direct costs of medical treatment [5]. The global cost to society per person with schizophrenia is $13,256 [10]. In 2020, the burden of mental disorders accounted for 25% of the total national burden of disease, topping the list [11]. Mental health expenditures as a percentage of total health care investment have been measured at an average of 2.8% globally, 5.1% in high-income countries, and 2.3% in China [12]. There is a need to improve the coverage of medical insurance for mental illness and give full play to the dual role of social insurance and commercial insurance. Therefore, it is crucial to scientifically predict hospitalization costs based on clinical characteristics and other factors to establish reasonable payment standards, which not only improve coverage of medical insurance but also guide rational health care and maximize the value of limited health resources. To accurately estimate the cost of treating a patient, it is essential to select suitable variables and optimize prediction models [13].

Over the decades, global researchers have identified critical factors that significantly influence the costs of patients with mental disorders. These factors included demographics (age), clinical characteristics (primary diagnosis, secondary diagnosis, physical impairment/activities of daily living, disease severity [14], depression, cognitive deficits, and aggressive or disruptive behaviors [15]), care utilization (location, attribute, and nature of settings, length of hospital stay, and accident and emergency), treatment (special medical procedures, nursing intensity, 1:1 monitoring), and social factors (social support [16], legal status, and referral source [17–21]. These variables are all potential factors to improve the predictive and explanatory power of payment models for inpatients with mental disorders. Currently, machine learning models have demonstrated promising results in predicting inpatient costs and exhibit strong potential for application, outperforming traditional regression models [22–24]. Researchers worldwide have studied and explained the use of machine learning methods for predicting inpatient costs, revealing high levels of accuracy. These models are capable of analyzing vast amounts of clinical, demographic, and healthcare cost data to make sensitive, flexible, and detailed predictions [25]. In global research, various machine learning algorithms, including least absolute shrinkage and selection operator (LASSO), ridge, classification and regression trees (CART), random forest (RF), support vector machine, and extreme gradient boosting (XGBoost), have been utilized to predict inpatient costs [26–30]. And there are few studies applying machine learning for cost prediction of patients with mental illness worldwide, indicating significant untapped potential [31, 32]. Traditional methods such as auto regressive integrated moving average (ARIMA) [33], weighted least squares regression modeling [34], and regression estimation [35] are commonly used in research on predicting mental health care costs, with limited utilization of machine learning methods. Among the limited studies that have employed machine learning to predict mental health costs, most scientific studies on predictive modeling have used categorical dependent variables for clustering levels of healthcare resource use due to skewed distributions, such as studies that typically focus on predictions for “high-cost” groups [36–39]. Researches on machine learning prediction of resource utilization of mental healthcare with continuous dependent variables have often lacked comparisons across different machine learning methods [40–42], with predictions for other healthcare resources (e.g., length of stay [43], number of visits [44], cost-effectiveness [40]); or with patients confined to a particular condition, such as depression [45], schizophrenia or schizoaffective disorder [46], and adolescents with intellectual and developmental disabilities [47].

In order to improve the prediction of hospitalization costs for patients with mental disorders and facilitate payment reform in China, this study utilized data from Jinhua city, a southeastern city in Zhejiang Province of China and established models based on machine learning methods. Jinhua is a pioneer city in establishing a refined per-diem payment for post-acute cases. It collects patient clinical and functional data with comprehensive assessment scales which enables advanced study in cost prediction. Since the cost per-diem is relatively stable for inpatients with mental disorders, this study constructed machine learning models to predict average daily hospitalization costs (ADHC) via patients’

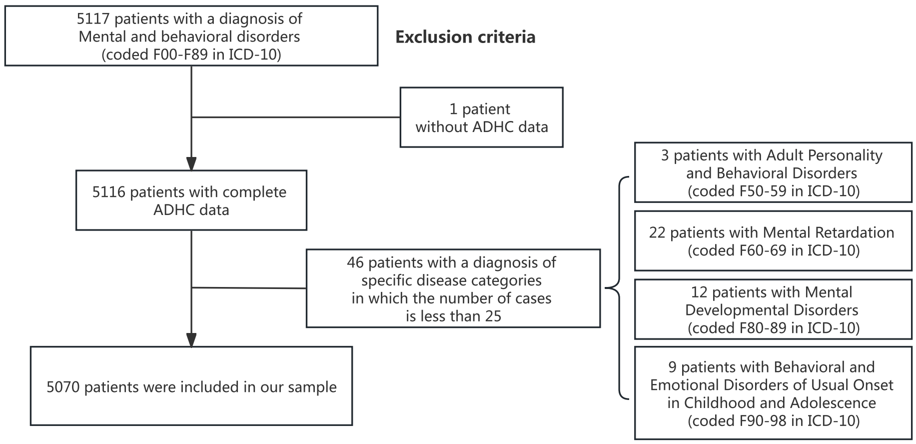

Fig. 1 Data processing and exclusion criteria

demographic, clinical, and functional variables, which is the first large-sample study in China to predict the hospitalization cost of mental disorders with comprehensive patient characteristics.

Methods

Data source and preprocessing

This study collected a total of 5,117 patient cases of mental disorders from 46 hospitals in Jinhua city, Zhejiang Province, Southeast of China, between January 1, 2022, and December 31, 2022. The sample selection criteria included inpatients with mental disorders classified under International Classification of Diseases (ICD) F00-F98. The inclusion and exclusion criteria for the samples were as follows: conformity with the ICD-10 criteria for mental and behavioral disorders; completeness of assessment scales, medical record homepages, and medical insurance claims information, with the ability to match patients on the basis of hospitalization serial number. Information on the total cost and institutional level was collected from the claims database of the Jinhua Healthcare Security Administration. Patient characteristic data were collected from the electronic information system for clinical condition assessment maintained by the Jinhua Healthcare Security Administration.

Within the dataset of 5,117 patients, one case with a missing value for ADHC was excluded. Additionally, diagnostic categories with fewer than 25 samples in the primary diagnosis were also excluded, including F50-F59: adult personality and behavioral disorders (3 cases), F60-F69: mental retardation (22 cases), F80-F89: psychological developmental disorders (12 cases), and F90-F98: behavioral and emotional disorders usually arising in childhood and adolescence (9 cases) (Fig. 1).

Candidate predictors

Demographic variables

Age, sex, and type of medical insurance, with the latter categorized into Urban Employee Basic Medical Insurance and Urban and Rural Resident Basic Medical Insurance.

Clinical and functional characteristics variables

Primary diagnosis, cognitive ability classification, depression, functional ability classification, and classification of comorbidity count.

Primary diagnosis category The primary diagnosis is categorized under mental and behavioral disorders (F00-F99) according to the International Statistical Classification of Diseases and Related Health Problems (10th Revision), with the first two digits of the code dividing the conditions into six major categories: F0 (F00-F09: organic, including symptomatic, mental disorders), F1(F10-F19: mental and behavioral disorders due to psychoactive substance use), F2(F20-F29: schizophrenia, schizotypal, and delusional disorders), F3(F30-F39: mood (affective) disorders), F4(F40-F49: neurotic, stress-related, and somatoform disorders), F7(F70-F79: mental retardation). These diagnoses are made by professional clinical physicians and medical staff across various medical institutions and are collected in the medical insurance bureau's patient information database.

Activities of Daily Living (ADL) grading The ADL is assessed via the Barthel Index [48], a widely utilized tool developed by Barthel and colleagues for evaluating daily living activities. The index includes ten items: eating, grooming, bathing, dressing, toilet use, transferring, walking, climbing stairs, bowel control, and bladder control. The total score ranges from 0 (complete dependence) to 100 (complete independence). ADL functional status is categorized into four levels based on the total score of the Barthel Index: 100–60 points indicate complete independence and self-care in daily living; 60–40 points suggest mild dependence with the need for assistance in daily activities; 40–20 points indicate moderate dependence, requiring significant help in daily activities; and scores below 20 points denote complete dependence with total reliance on others.

Cognitive grading The Chinese version of the Continuity Assessment Record and Evaluation (CARE) is used to assess cognitive function [49]. The scale has been validated for reliability and localization [50]. The scale consists of 11 items evaluating several dimensions, including mental status, temporal orientation, and memory. The total score ranges from 0 to 15, with lower scores indicating more severe cognitive impairment. 13 to 15 indicates normal cognitive functioning, whereas scores below 13 indicate cognitive impairment, with mild cognitive impairment defined as a score of 8 to 12, moderate cognitive impairment as a score of 1 to 7, and severe cognitive impairment as a score of 0.

Depression status The Chinese version of CARE is also used to assess the patients' depressive status [50]. It encompasses common symptoms of depression, such as low mood, diminished interest, sleep disturbances, altered appetite, poor self-esteem, and difficulties with concentration. The patient is qualified as depressed if the total score is greater than or equal to 10.

Grading of the number of comorbidities The grading is based on the number of comorbidities present: 0 indicates no comorbidities; 1 indicates one comorbidity; 2 indicates two or three comorbidities; and 3 indicates four or more comorbidities.

Institutional variables

The level of medical institution is categorized into three tiers: tertiary medical institutions, secondary medical institutions, and non-graded medical institutions.

Outcome variable

For inpatients with mental disorders, ADHC serves as a common and robust indicator for evaluating the expenditure associated with psychiatric hospitalization. ADHC is calculated by dividing the total hospitalization costs by the number of hospitalization days and is utilized as the dependent variable in ADHC prediction model.

In this study, demographic information, clinical characteristics, and institutional data encompassing 9 variables are selected as explanatory variables for prediction (Table 1), which aligns well with the literature [43, 51–53]. The presence of comorbidities are related to the classification of the number of comorbidities, leading to potential issue of multicollinearity, so we didn't include the "the presence of comorbidities" variable in the model.

Modeling

Initially, correlation and multiple regression analyses were conducted on the independent and dependent variables to evaluate the relationships between the logarithm of ADHC and various characteristic variables. For the correlation analysis, the Spearman correlation method was selected, and a correlation matrix encompassing all features was produced (Fig. 2). The multiple regression analysis was based on the ordinary least squares (OLS) regression method, with results expressed as regression coefficients and their corresponding 95% confidence intervals (CIs). For the primary diagnosis, dummy variable treatment was applied via OneHotEncoder, with F0 serving as the reference category.

Subsequently, predictive models were constructed via both traditional linear regression and machine learning models. In the machine learning models, variables that were not significant in the univariate analysis were not omitted. For instance, the cognitive classification variable, which demonstrated significant performance in the multiple regression model, was retained.

This study constructed predictive models via linear regression and five machine learning models, including generalized linear regression (GLR), support vector regression (SVR), LASSO regression, CART, RF, and XGBoost, to model the prediction of patients' ADHC. International studies have often favored linear regression models as a benchmark [25, 54–56]. By comparing the predictive performance of machine learning models, this study contributes to international comparative research in the field. GLR represents a generalization of linear models. LASSO mitigates the impact of multicollinearity. SVR falls under the umbrella of support vector machine models, and CART constitute a family of decision tree models. Furthermore, RF and XGBoost are ensemble models that utilize distinct integration techniques—bagging and boosting, respectively. Employing a diverse array of models may enhance the accuracy of ADHC predictions. 80% of the data was used for training and 20% of the data was used for testing. The hyper-parameters of all models were fine-tuned using a 5-fold cross-validation combined with grid search on the training datasets [57, 58]. The performance differences among the models were computed utilizing a bootstrapping method with 1000 iterations. ADHC prediction models were developed and evaluated in a Windows operating system environment via Python (version 3.12.5) scikit-learn, xgboost packages.

Table 1 Statistical characteristics of 5,070 inpatients with mental disorders

Variables	Feature	N(%)	Univariable analysis		Multivariable analysis	
			Correlation coefficient	*P*-value	Correlation coefficient (95%CI)	*P*-value
Age			0.0674	0.000*	-0.0020(-0.0026,-0.0013)	0.000*
	Age ≤ 18	67(1.32%)				
	18 < Age ≤ 60	3502(69.07%)				
	60 < Age ≤ 85	1252(24.69%)				
	85 < Age	249(4.91%)				
Sex			-0.0955	0.000*	-0.0144(-0.0322,0.0033)	0.111
	1: Male	3033(59.82%)				
	0: Female	2037(40.18%)				
Type of medical insurance			0.2528	0.000*	0.0588(0.0358,0.0818)	0.000*
	1: Urban Employee Basic Medical Insurance	936(18.46%)				
	0: Urban and Rural Resident Basic Medical Insurance	4134(81.54%)				
Level of medical institutions			0.6504	0.000*	0.2557(0.2455,0.2660)	0.000*
	1: Non-graded medical institutions	1805(35.60%)	-0.5408	0.000*		
	2: Secondary medical institutions	1292(25.48%)	-0.0971	0.000*		
	3: Tertiary medical institutions	1973(38.92%)	0.6178	0.000*		
Primary diagnosis	Dummy variable processing		-0.0579	0.000*		
	F0	666(13.14%)	0.1803	0.000*	Reference	Reference
	F1	113(2.23%)	-0.0874	0.000*	-0.2927(-0.3569,-0.2284)	0.000*
	F2	3287(64.83%)	-0.1465	0.000*	-0.2790(-0.3111,-0.2468)	0.000*
	F3	513(10.12%)	0.1186	0.000*	-0.1210(-0.1633,-0.0786)	0.000*
	F4	73(1.44%)	0.0337	0.016**	-0.0739(-0.1502,0.0024)	0.058
	F7	418(8.24%)	-0.0648	0.000*	-0.3255(-0.3702,-0.2809)	0.000*
Whether depression			0.0595	0.000*	0.0418(0.0206,0.0631)	0.000*
	1: Yes	1033(20.37%)				
	0:No	4037(79.63%)				
Functional classification			-0.0339	0.016**	-0.0466(-0.0598,-0.0334)	0.000*
	1: Complete dependence	466(9.19%)	0.2361	0.000*		
	2: Moderate dependence	212(4.18%)	0.0062	0.660		
	3: Mild dependence	2535(50.00%)	-0.2059	0.000*		
	4: Complete independence	1857(36.63%)	0.0695	0.000*		
Cognitive classification			0.0269	0.056	-0.0369(-0.0489,-0.0249)	0.000*
	1: Severe cognitive impairment	78(1.54%)	0.1263	0.000*		
	2: Moderate cognitive impairment	2344(46.23%)	-0.0402	0.004*		
	3: Mild cognitive impairment	1182(23.31%)	-0.0741	0.000*		
	4: Normal cognitive function	1466(28.92%)	0.0790	0.000*		
Classification of comorbidity count			0.0829	0.000*	0.0067(-0.0021,0.0155)	0.135
	0: No comorbidity	1118(22.05%)	0.0227	0.105		
	1: One comorbidity	1293(25.50%)	-0.0842	0.000*		
	2: Two or three comorbidities	1483(29.25%)	-0.0902	0.000*		
	3: Four or more comorbidities	1176(23.20%)	0.1618	0.000*		

F0 F00-F09: organic, including symptomatic, mental disorders; F1 F10-F19: mental and behavioral disorders due to psychoactive substance use; F2 F20-F29: schizophrenia, schizotypal, and delusional disorders; F3 F30-F39: mood (affective) disorders; F4 F40-F49: neurotic, stress-related, and somatoform disorders; F7 F70-F79: mental retardation. All categorical variables except age, which is a continuous variable

* $P < 0.01$

** $P < 0.05$

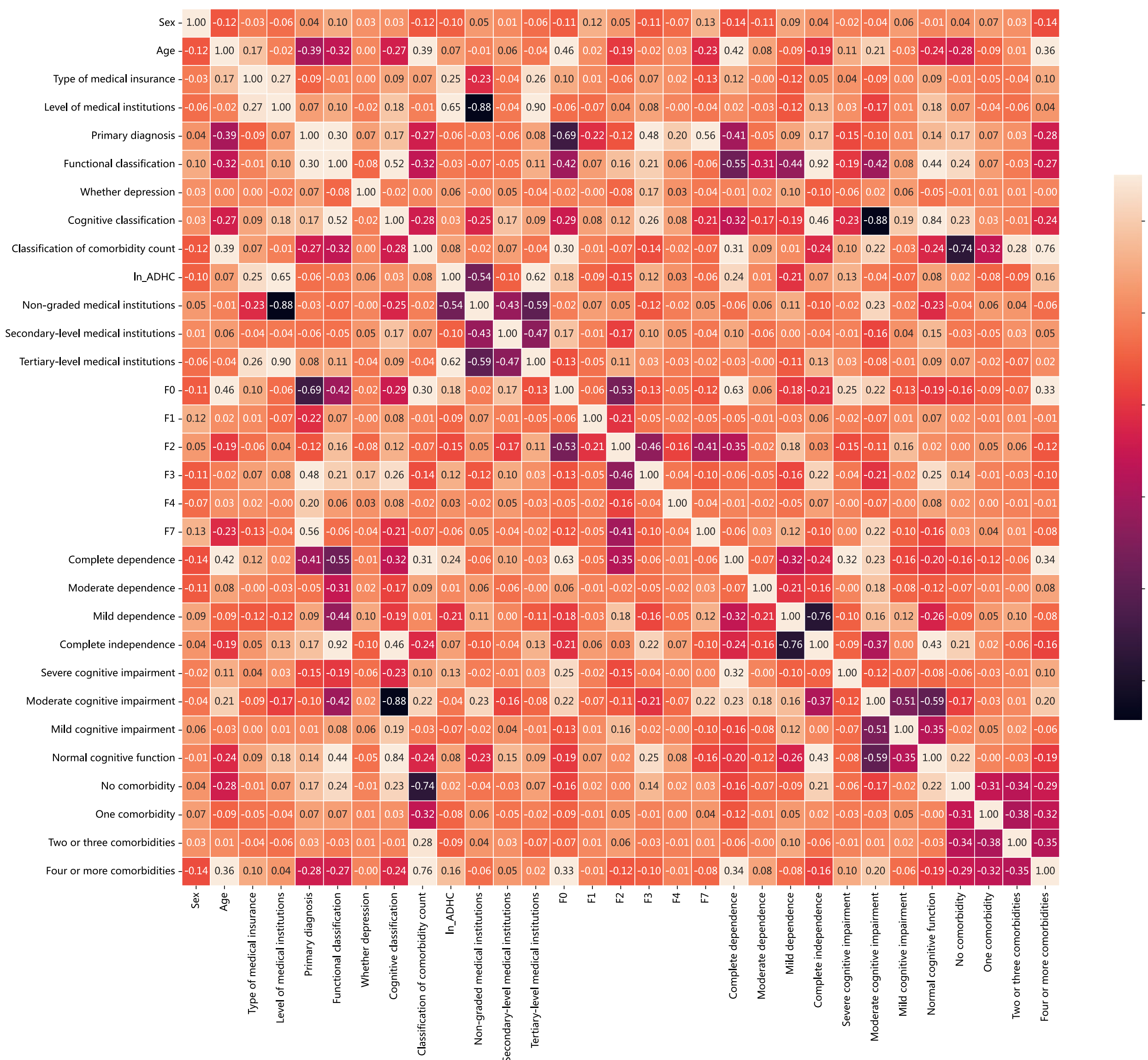

Fig. 2 Correlation matrix for all features in the dataset. Strong positive correlation is represented by white, strong negative correlation by black, and the orange color denotes a weak correlation. F0 F00-F09: organic, including symptomatic, mental disorders; F1 F10-F19: mental and behavioral disorders due to psychoactive substance use; F2 F20-F29: schizophrenia, schizotypal, and delusional disorders; F3 F30-F39: mood (affective) disorders; F4 F40-F49: neurotic, stress-related, and somatoform disorders; F7 F70-F79: mental retardation

GLR: Multivariate linear regression is employed to elucidate the relationships among the dependent variable, the logarithm of ADHC, and the explanatory variables. The resulting multivariate regression model is defined as $y = \beta_0 + \sum_{i=1}^{n} \beta_i x_i$. The inclusion of dummy variables may influence the fit of the GLR. For example, in the calculation of ADHC for patients with mental disorders, it is challenging to incorporate various disease groups as parameters within the model, despite their significant impact on medical costs. Therefore, machine learning algorithms are considered to address issues that traditional algorithms cannot resolve. Parameters: fit_intercept = True, copy_X = True, n_jobs = None.

LASSO regression is a regularized linear model that reduces model complexity, effectively addressing multicollinearity in regression models, and guarding against overfitting [59]. Parameter: alpha = 0.0001.

The SVM algorithm, proposed by Cortes and Vapnik, constructs a hyperplane in high-dimensional space for both classification and regression tasks [60]. SVR is one such application of SVM. The model is expressed as $y = \sum_{i=1}^{n} \left(\alpha_i - \alpha_i^*\right) K\left(x_i, x_j\right) + b$, where x represents the explanatory variables in multivariate linear regression, y is the logarithm of ADHC, α_i and α_i^* are Lagrange multipliers, b is a constant, and K is the kernel function. Parameters: C = 15, epsilon = 0.15, gamma = 0.01, kernel = 'rbf'.

CART utilizes decision trees to recursively partition data based on feature values [61]. Parameters: max_depth = 8, min_samples_leaf = 7, random_state = 42, splitter = 'random'.

RF, proposed by Breiman, is an ensemble learning method based on decision trees [62]. It can handle high-dimensional features, prevent overfitting, and is particularly adept at managing large-scale datasets, yielding satisfactory training outcomes for both discrete and continuous data types. The algorithm can output a ranking of influential factors, assessing the role of each variable in regression. Parameters: max_depth = 10, max_features = 5, min_samples_leaf = 3, n_estimators = 45.

XGBoost is an optimized implementation of gradient-boosted decision trees designed for speed and performance, as proposed by Chen and Guestrin [63]. The algorithm is optimized in terms of computational resource utilization. It employs a gradient-boosting framework and is also an ensemble method based on decision trees. However, it uses an enhanced approach where each tree aims to minimize the errors of the preceding tree and improves the objective optimization function by optimizing the loss function and complexity penalty. It can output a ranking of feature importance. Parameters: colsample_bytree = 0.8, gamma = 0, learning_rate = 0.15, max_depth = 6, min_child_weight = 0.1, n_estimators = 100, objective = 'reg:squaredlogerror', reg_alpha = 0.05, reg_lambda = 0.01, scale_pos_weight = 0.1.

Results

Demographic and clinical features of the sample

Owing to the right-skewed, long-tailed characteristics of ADHC data, logarithmic transformation is necessary for model construction. After data preprocessing, a total of 5,070 cases were included in the study (Table 1). The distribution of the logarithmically transformed ADHC data was fitted via the Python Fitter package, which was found to conform to either a logistic or normal distribution.

The primary diagnostic categories included in the study are as follows: F0, F1, F2, F3, F4, and F7.

As depicted in Table 1, variables are expressed in absolute frequency and percentage (N, %) and were subjected to both correlation and multiple regression analyses [64], with a correlation matrix including all features being output [65, 66] (Fig. 2). The results of the multiple regression analysis are presented as regression coefficients along with their corresponding 95% CIs. The model intercept was 5.6567 (95% CI: 5.585–5.729); R-squared (R^2) was 0.414, with an adjusted R^2 of 0.413; the mean squared error (MSE) was 25.1764; the residual MSE was 0.0916; the overall MSE was 0.1559; the regression sum of squares (SSR) was 463.1822; the Akaike information criterion (AIC) was 2284; the Bayesian information criterion (BIC) was 2375; the F-statistic was 275; and the F-statistic p-value was 0.000. A two-tailed p-value of less than 0.01 indicates a statistically significant difference.

Performance of the machine learning models

Different ADHC prediction models excel in various application scenarios and prediction aspects. Therefore, the study evaluated the predictive performance of each model using R^2, root-mean-square error (RMSE), mean-absolute error (MAE), and mean-absolute-percentage error (MAPE) [25, 56, 67, 68]. R^2 is used to assess the model's fitting ability, while RMSE, MAE, and MAPE measure the prediction error. A higher R^2 value closer to 1 indicates better model fitting, while lower RMSE, MAE, and MAPE values suggest reduced prediction errors.

The expressions for R^2, RMSE, MAE, and MAPE are as follows: where $\widehat{y}_i$ represents the predicted value, y_i represents the test value, and $\bar{y}$ represents the mean of the test values.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \widehat{y}_i\right|$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \widehat{y}_i}{y_i}\right|$$

The performance of six model predictions was quantified via the aforementioned evaluation parameters, and difference test between models' R^2, which were evaluated through the average performance of the bootstrapping method with 1000 repetitions [69].

Table 2 presents the average values of R^2, RMSE, MAE, and MAPE for GLR, LASSO, SVR, CART, XGBoost, and RF, which are based on the bootstrap method. Supplementary Fig. 1 presents a comparison of actual versus predicted values across various models. RF model emerged as the better performer, achieving the highest R^2 of 0.6417 (95% CI, 0.6236–0.6611), an improvement of 58% compared to the GLR. It also exhibited the lower RMSE, MAE, and MAPE values of 0.2398 (95% CI, 0.2252–0.2553),

Table 2 Predictive performance of models. Values are the average of resampling 1000 times by bootstrap method. Difference test of the R^2 between machine learning models and GLR (the reference model) using the bootstrap method

Model	R^2	*P*-value	RMSE	MAE	MAPE
GLR	0.4062(0.3821–0.4336)	[Reference]	0.3088(0.2864–0.3319)	0.2152(0.2084–0.2222)	0.0380(0.0369–0.0391)
LASSO	0.4062(0.3821–0.4336)	0.5490	0.3088(0.2864, 0.3319)	0.2152(0.2083–0.2221)	0.0380(0.0369–0.0390)
SVR	0.5445(0.5207–0.5713)	0.0000	0.2705(0.2512–0.2915)	0.1859(0.1803–0.1926)	0.0326(0.0318–0.0336)
CART	0.5576(0.5318–0.5849)	0.0000	0.2665(0.2512–0.2832)	0.1844(0.1786–0.1906)	0.0324(0.0315–0.0334)
XGBoost	0.6095(0.5904–0.6300)	0.0000	0.2504(0.2356–0.2666)	0.1742(0.1689–0.1800)	0.0306(0.0297–0.0315)
RF	0.6417(0.6236–0.6611)	0.0000	0.2398(0.2252–0.2553)	0.1677(0.1626–0.1735)	0.0295(0.0287–0.0304)

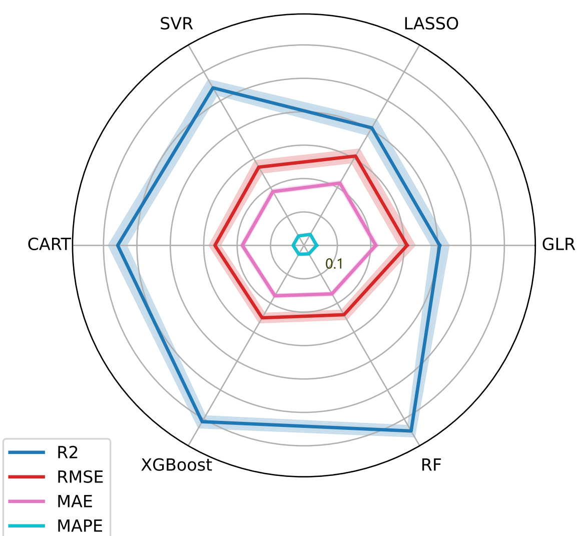

Fig. 3 Radar graph showing the predictive performance of models

0.1677 (95% CI, 0.1626–0.1735), and 0.0295 (95% CI, 0.0287–0.0304), respectively, indicating strong generalizability. XGBoost improved the R^2 by 50%, reaching 0.6095 (95% CI, 0.5904–0.6300) compared to the GLR, followed by CART, SVR, LASSO, and GLR model. RF, XGBoost, CART and SVR are statistically significant relative to GLR based on difference tests (p-value < 0.01).

In this study, the GLR and LASSO models presented approximate R^2, RMSE, MAE, and MAPE values.

Model selection analysis

Selecting the most effective machine learning model is a complex task. This process involves considering numerous performance parameters while weighing the insights derived from the results and their relevance to the clinical domain. Table 2 and Fig. 3 confirm the superiority of RF models, as R^2 value is relatively high, while RMSE, MAE, and MAPE values are lower, indicating consistency and stability across the resampling. Based on their feature importance scores (Fig. 4), we chose the top 4 (feature importance > 0.05) and top 11 features (feature importance > 0.01) for inclusion in the model. Table 3 provides a detailed breakdown of the R^2, RMSE, MAE, and MAPE values for RF models. The findings reveal that models integrating the initial four features exhibit inferior performance. Notably, the R^2 value of the model incorporating all features marginally exceeds that of the model that includes the top 11 features. Nevertheless, when assessed using RMSE, MAE, and MAPE, the latter model demonstrates a slight edge over the former.

The actual dataset contains a number of outlier data points attributable to individual heterogeneity, which encompass patients with both low and abnormally high ADHC. IsolationForest is a highly robust anomaly detection algorithm that employs a unique tree-based framework to effectively isolate outliers [70]. So, we employed the IsolationForest algorithm to handle outliers in the logarithmic ADHC (Sup. Figure 2). However, after removing 1% (49 entries) of the outlier data from the entire dataset, we found that the R^2 of RF and XGBoost models, as shown in Table 4, improved by only 6% and 11%, respectively. Given the strong research value of this subset of data, further exploration through population stratification studies could lead to new breakthroughs in predicting ADHC of patients with mental disorders and enhancing medical quality and efficiency [71]. Therefore, we may not recommend using the IsolationForest algorithm to handle outliers in the logarithmic ADHC.

Feature importance

Four models— RF, XGBoost, CART, and SVR, were selected to identify four sets of key predictors, thereby elucidating the factors influencing ADHC for patients with mental disorders. Significant predictors were ranked in descending order of their importance in predicting the outcome (Fig. 4). Predictor importance was determined by its impact on the model's predictive performance, with the most influential predictor at the top.

Based on the feature importance derived from RF model and considerations from the other three models,

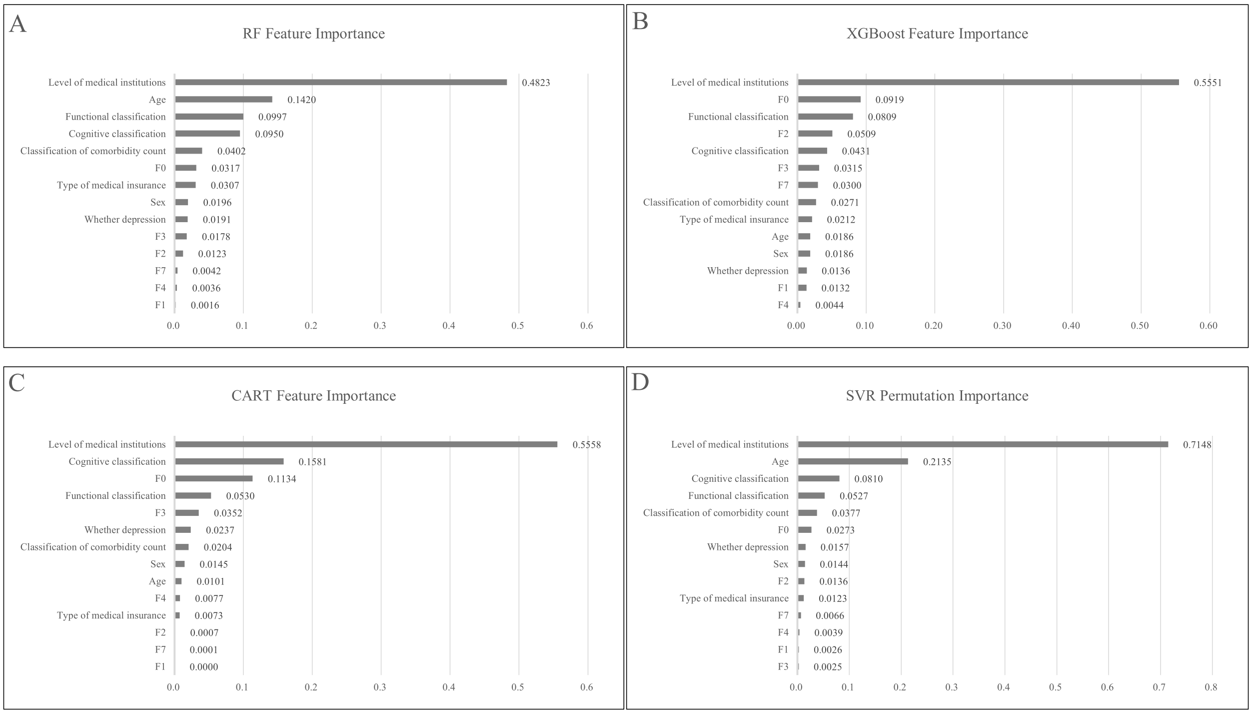

Fig. 4 Importance of each variable in RF, XGBoost, CART and SVR models. **A** The RF model. **B** The XGBoost model. **C** The CART model. **D** The SVR model. The feature importance is a measure scaled to have a maximum value of 1

features were categorized into three groups: strong, moderate, and weak influence.

Features with strong influence (feature importance > 0.05, ranked 1–4) include level of medical institutions, age, functional classification, and cognitive classification. Notably, level of medical institutions was the most significant factor across all four models, consistent with its prominent role in both univariate and multivariate analyses (Table 1), where the significance level was $p < 0.01$. Functional classification also appeared among the top four influential factors in all four models, whereas cognitive classification was present in three models and age in two models.

Factors with moderate influence (feature importance between 0.01 and 0.05, ranked 5–11) include classification of comorbidity count, F0, type of medical insurance, sex, depression status, F3 and F2. Among these, classification of comorbidity count, type of medical insurance, and sex were consistently ranked as moderately influential across all four models. Depression status and F3 appeared in three models, while F0 and F2 were present in two models.

Features with weak influence (feature importance < 0.01, ranked 12–14) include F7, F4 and F1. F1 appeared as weakly influential factors in all four models, F4 and F7 in three and two models, respectively.

Overall, the high concentration of model feature categories provides a solid foundation for analyzing the primary factors influencing ADHC.

Visualizing statistical analysis

The centered individual conditional expectation (cICE) [72] chart illustrates the primary factors influencing ADHC (Fig. 5A-D) and presents interaction plots. The

Table 3 Predictive performance of the RF model with the top 4(feature importance > 0.05) and top 11(feature importance > 0.01) features

RF Model	R^2	RMSE	MAE	MAPE
top 4 features	0.5459 (0.5244–0.5680)	0.2362 (0.2294–0.2433)	0.1803 (0.1756–0.1855)	0.0322 (0.0313–0.0331)
Top 11 features	0.6413 (0.6243–0.6582)	0.2100 (0.2043–0.2162)	0.1621 (0.1581–0.1663)	0.0290 (0.0283–0.0297)

The feature importance values of the top 4 and top 11 features are greater than 0.05 and 0.01 respectively. Values are the average of resampling 1000 times by bootstrap method

Table 4 Predictive performance of the RF and XGBoost models with 1% outlier data deleted

Model	R^2	RMSE	MAE	MAPE
XGBoost	0.6773 (0.6589–0.6944)	0.1991 (0.1935–0.2052)	0.1516 (0.1477–0.1555)	0.0271 (0.0264–0.0277)
RF	0.6782 (0.6609–0.6953)	0.1988 (0.1931–0.2049)	0.1525 (0.1484–0.1564)	0.0272 (0.0265–0.0279)

The RF and XGBoost models performed better, and these two models were used as reference. Values are the average of resampling 1000 times by the bootstrap method

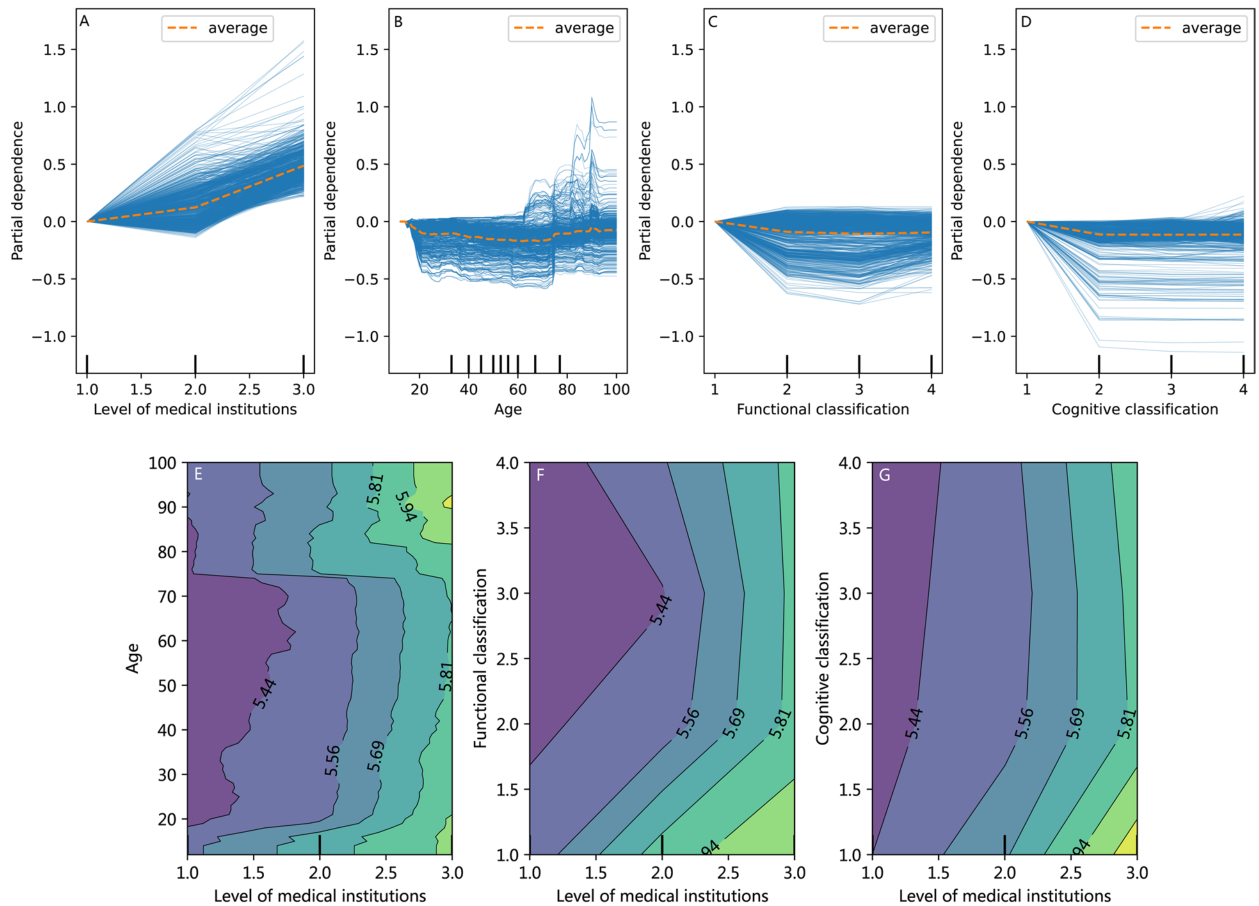

Fig. 5 Top 4 important features centered individual conditional expectation (cICE) and 2-way interaction plots with RF. **A**–**D** The top 4 features. Partial dependence interprets ADHC response as a function of the top 4 features of interest. The y-axis is interpreted as the predicted change relative to the baseline value or the leftmost value. **E**–**G** Cross-analysis of medical institutions' level with age, functional classification, and cognitive classification

heat matrix corresponds to the magnitude of the logarithm of ADHC, with deeper purple indicating lower ADHC and lighter yellow signifying higher AHDC.

The ICE charts reveal that, generally, patients admitted to higher-level medical institutions (Fig. 5A; Sup. Figure 3A), those aged below 20 and above 75 (Fig. 5B; Sup. Figure 3B), those with lower functional classification indicating higher dependency (Fig. 5C; Sup. Figure 3C), and those with lower cognitive classification reflecting greater cognitive impairment (Fig. 5D; Sup. Figure 3D) tend to incur higher ADHC. Notably, the functional and cognitive classification curves exhibit a skewed V-shape, with significantly higher ADHC at the functional classification of grade 1 and cognitive classification of grade 1, exceeding 330 yuan and 322 yuan respectively, and a tailing effect at the functional classification of grade 4 and cognitive classification of grade 4.

A cross-analysis of medical institutions' level with age, functional classification, and cognitive classification

(Fig. 5E-G) reveals a significant nonlinear relationship between these factors and ADHC.

Overall, patients aged 20–75 years incur lower ADHC across all medical institutions' level compared to other age groups. Specifically, patients in this age range have lower ADHC in primary care institutions. ADHC for 20–75-year-old patients in secondary medical institutions remains relatively consistent. The prevalence of ADHC was notably higher in tertiary medical institutions, particularly among patients aged 85–100 years and those younger than 20 (Fig. 5E).

Patients with a functional classification of grade 1 have significantly higher ADHC than those with grades 2, 3, and 4. Among them, patients with the functional classification of grade 3 experience a relatively smaller increase in ADHC as higher medical institutions' level within and below level 2 compared to other grades, while grade 1 exhibits the opposite trend(Fig. 5F).

Similarly, patients with a cognitive classification of grade 1 have significantly higher ADHC than those with grades 2, 3, and 4. Patients with cognitive classification of grades 2, 3, and 4 display minimal variation in ADHC in secondary medical institutions compared to other grades. However, patients with the cognitive classification of grade 1 exhibit a significant increase in ADHC in tertiary medical institutions (Fig. 5G).

Discussion

The study identified four key variables crucial for predicting ADHC for patients with mental disorders: level of medical institutions, age, functional classification, and cognitive classification. As medical data science progresses, predicting ADHC for patients with mental disorders remains a challenge. This machine learning model, which incorporates basic patient information and scale measurements upon admission, enhances the convenience of prediction.

The level of medical institutions stands out as the most influential factor in the prediction model for ADHC, surpassing all other factors (Fig. 4A). Prior research has also highlighted the pivotal role of medical institutions' level in ADHC forecasting. Results from the Spearman correlation analysis (0.6504, p-value = 0.000), multivariate regression analysis (0.2557, p-value = 0.000), as well as machine learning models such as RF (feature importance = 0.4823), XGBoost (feature importance = 0.5551), CART (feature importance = 0.5558), and SVR (permutation importance = 0.7148), all underscore the significant impact of medical institutions' level on the predictive outcome. The reasons are threefold: firstly, higher-grade medical institutions offer a more comprehensive range of nursing and rehabilitation care options to meet the needs of inpatients with mental disorders; secondly, these hospitals charge higher fees for these packages, directly driving up ADHC; thirdly, higher-grade medical institutions predominantly treat patients with severe illnesses such as F0, F1, F2, F3, which are more costly. Tertiary medical institutions treat 91.28% of these patients, a higher percentage than secondary medical institutions (91.10%) and non-graded medical institutions (88.7%). Research by Li J, et al. (2023) [52] on inpatient costs for patients with schizophrenia in Gansu Province, China, also suggests that the average costs at district and county-level medical institutions are lower than those at provincial and municipal levels. Similarly, Ma et al. (2024) [73] reported that provincial-level medical institutions positively influence outpatient costs for depression in Liaoning, China, while primary medical institutions have a negative impact.

Age is another significant factor influencing ADHC (Fig. 4A). As a continuous variable in the model, age ranks as the second most important factor in both RF model (feature importance = 0.1420) and SVR model (permutation importance = 0.2135). Patients under 20 years old or over 75 years old with mental disorders have ADHC, averaging over 291 yuan (Sup. Figure 3B). This is partly due to the "Summary of Medical Service Prices in Provincial Public Hospitals of Zhejiang Province" (July 2021), which shows a 30% surcharge on the original price for "psychiatric nursing" and for children aged six and below, leading to increased ADHC. Additionally, patients over 75 years old have a 76.29% prevalence of F0, indicating a higher probability of severe diagnoses and consequently higher ADHC. Thirdly, elderly and young patients are less common, representing only 12.98% of the total population, which means that outliers may disproportionately affect the overall estimate. Cromwell et al. (2005) [51] suggested that age was a significant variable affecting routine costs for patients in U.S. psychiatric hospitals. Jones et al. (2007) [74], in a review of previous research on the costs of mental health care, reported that the impact of age on costs is complex, with older age potentially increasing or decreasing costs, and the relationship is not simply linear. Li J, et al. (2023) [52] identified age as one of the top five factors affecting annual average hospitalization cost for patients with schizophrenia in Gansu Province, China, and reported a negative correlation with the average cost per hospital stay.

Functional classification is yet another significant factor influencing ADHC (Fig. 4A). It is an important determinant in major machine learning models such as RF (Feature Importance = 0.0997), XGBoost (feature importance = 0.0809), CART (feature importance = 0.0530), and SVR (permutation importance = 0.0527). ADHC for different functional classifications are ranked as follows:

complete dependence (grade 1) > moderate dependence (grade 2) > complete independence (grade 4) > mild dependence (grade 3) (Sup. Figure 3C). Cromwell et al. (2005) [51] found that patients with three or more ADLs had a 43% increase in treatment intensity, and the length of stay curve also exhibited an inclined V-shape. For inpatients with mental disorders in the complete dependence functional classification, their pharmaceutical and nursing costs are much higher than those in other functional classifications, resulting in higher ADHC. Moderately dependent patients have slightly higher ADHC than those who are completely independent. Patients with complete independence in functional classification, often require sedative medication to stabilize their emotions, hence ADHC is higher than those with mild dependence, with the latter having the lowest ADHC.

Cognitive classification significantly impacts ADHC (Fig. 4A), emerging as a crucial factor across various models, including RF (feature importance = 0.0950), CART (feature importance = 0.1581), and SVR (permutation importance = 0.0810). ADHC follows a specific pattern: severe cognitive impairment (grade 1) > normal cognitive function (grade 4) > mild cognitive impairment (grade 3) > moderate cognitive impairment (grade 2) (Sup. Figure 3D). Cromwell et al. (2005) [51] observed a 24% increase in treatment intensity for patients with cognitive impairments. Notably, patients with severe cognitive impairment incur significantly higher costs for medications and nursing, leading to higher ADHC. Conversely, patients with normal cognitive function may have higher drug costs, corroborating findings from functional classification among completely independent patients.

Apart from the above factors, classification of comorbidity count is another significant factor influencing ADHC (Fig. 4A). Prince et al. (2007) [75] postulated that comorbidities complicate the diagnosis and treatment of mental disorders, thereby affecting prognosis. This factor holds substantial importance in RF (feature importance = 0.0402). ADHC increases with the number of comorbidities: four or more comorbidities > two or three comorbidities > one comorbidity > no comorbidity (Sup. Figure 3E). The presence of comorbidities leads to an increase in medications, nursing, and rehabilitation options, ultimately driving up ADHC. Li J, et al. (2023) [52] echoed this finding in their study on hospitalization costs for schizophrenia in Gansu Province, China, while Dai, et al. (2023) [53] focused on elderly patients with F0, F1, F2, F3, F4, F7 mental disorders and found that comorbid physical illnesses could exacerbate mental symptoms, indirectly increasing ADHC.

The parameter estimates of the primary diagnosis also exert a considerable influence on RF model's outcomes (Fig. 4A). The primary diagnosis, treated as a dummy variable, shows significant importance for conditions such as F0 (feature importance = 0.0317), F2 (feature importance = 0.0123), and F3 (feature importance = 0.0178). Based on multivariate analysis, with F0 as the reference, the coefficients for F1, F2 and F3 were -0.2927 (p-value = 0.000), -0.2790 (p-value = 0.000), and -0.1210 (p-value = 0.000), respectively. This finding indicates a positive correlation between the primary diagnosis and ADHC, suggesting an overall positive impact on costs with increasing severity of diagnosis. Cromwell et al. (2005) [51] reported that elderly patients with severe mental disorders incur higher costs, while Jones et al. (2007) [74] emphasized the clinical significance of diagnosis and severity as critical predictors of higher costs.

Patient sex plays a role in predicting ADHC (Fig. 4A). Our study revealed that female patients tend to have higher ADHC than male patients (Sup. Figure 3F). This may be attributed to the higher proportions of female patients in tertiary medical institutions (42.61%), complete dependency (14.09%), and those with four or more comorbidities (30.54%) compared to males (36.43%; 5.90%; 18.27%). Additionally, the average age of female patients (56.45 years) is higher than that of male patients (51.94 years), and they have a higher risk of severe diagnosis F0 (17.53% vs. 10.19%). Chisholm et al. (1997) [76] also reported lower costs for males than for females. Lu et al. (2021) [77] reported a higher prevalence of depression among women in China, highlighting the need for accessible treatment for female patients.

Analysis reveals that machine learning outperforms to the traditional GLR model in predicting ADHC for patients with mental disorders. Specifically, machine learning typically exhibits lower data distribution requirements, broader applicability, and higher efficiency than the conventional GLR model does. In detail, XGBoost significantly improves the fitting effect. However, its performance is slightly inferior to RF. CART models are faster in computation than RF, but their performance lags behind RF. SVR model demonstrates a notable improvement over linear models, with overall performance at a moderate level. LASSO regression performs well in addressing collinearity issues among variables in the dataset, but based on the findings of this study, the impact of collinearity among variables in the dataset is minimal.

Overall, RF model performs better than the other models in this study, achieving better fitting and generalization capabilities. Compared to traditional statistical methods, the RF model exhibits superior performance, making it suitable for capturing complex nonlinear relationships and robust to outliers. While its computation speed is slower than a single decision tree, controlling noisy predictor variables by pruning individual trees to their maximum depth can enhance the model's generalization ability [78]. The RF model's powerful predictive capabilities have been applied in clinical cost predictions and screenings, diagnosis, and treatment in medical research [54, 58, 65]. This study lays a foundation for further external validation and potential practical applications of the RF model, opening new avenues to predict ADHC for patients with mental disorders Additionally, the good fit of all models (R^2 greater than 0.4) provides high credibility for the accuracy of the model predictions.

In summary, the study employed various interpretable machine learning models to predict ADHC for patients with mental disorders. The RF model revealed key factors influencing ADHC for patients with mental disorders including medical institutions level, age, functional classification, cognitive classification, classification of comorbidity count, and primary diagnosis. The research demonstrated that machine learning algorithms, particularly RF algorithm, improved the accuracy of ADHC prediction.

Study limitations

Inevitably, this study has several limitations. Firstly, owing to data availability, the research has not incorporated clinical indicators specifically reflecting the severity of mental disorders into the predictive analysis, even though such indicators are known to significantly enhance the predictive efficacy of models in related studies. Future research could incorporate more specific indicators reflecting the severity of mental disorders. Secondly, the current sample data is primarily derived from the healthcare insurance system of a single prefecture-level city. Differences in economic development levels and healthcare systems could potentially lead to the non-transferability of the model. Nevertheless, given the excellent fitting and generalization capabilities of the machine learning model, the results of this study may still provide a reference for other regions.

Conclusions

The findings of this study aid in optimizing health insurance payment methods for hospitalization costs for patients with mental disorders and improve clinical practices. Although per-diem payment is commonly used for patients with mental disorders, the lack of scientific methods for cost prediction, in reality, hinders the establishment of reasonable payment standards. This is the first large-sample study in China to predict the hospitalization cost of mental disorders with comprehensive patient characteristics and advanced machine learning models. It is probable that these findings will facilitate the establishment of more reasonable insurance payment standards for inpatients with mental disorders. Possibly, given the continuous development of commercial insurance, the research results can also assist in determining payment standards for mental disorders hospitalizations, addressing the medical and income loss compensation needs of inpatients with mental disorders, and providing practical bases for product pricing and design. Additionally, in some cases, the findings may also enable medical institutions to reasonably implement clinical interventions for patients, thereby reducing the financial risks faced by these institutions and enhancing resource utilization efficiency.

Abbreviations

LASSO	Least absolute shrinkage and selection operator
CART	Classification and regression trees
RF	Random forest
XGBoost	Extreme gradient boosting
ARIMA	Auto regressive integrated moving average
ADHC	The average daily hospitalization costs
ICD	International Classification of Diseases
ADL	Activities of Daily Living
CARE	The Continuity Assessment Record and Evaluation
OLS	Ordinary least squares
F0	F00-F09 Organic, including symptomatic, mental disorders
F1	F10-F19 Mental and behavioral disorders due to psychoactive substance use
F2	F20-F29 Schizophrenia, schizotypal and delusional disorders
F3	F30-F39 Mood (affective) disorders
F4	F40-F48 Neurotic, stress-related and somatoform disorders
F7	F70-F79 Mental retardation
CIs	Confidence intervals
R^2	R-squared
MSE	Mean squared error
SSR	Regression sum of squares
AIC	Akaike information criterion
BIC	Bayesian information criterion
RMSE	Root-mean-square error
MAE	Mean-absolute Error
MAPE	Mean-absolute-percentage error
GLR	Generalized linear regression
SVR	Support vector regression
cICE	Centered individual conditional expectation

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

All authors contributed to the concept and outline of the manuscript. L. H. and Y. M. conceived and designed the study. X. L. was involved in the data collection, conducting preliminary data organization. X. T. contributed to the literature review and the raw data cleaning. Y. M. performed the statistical analyses and the interpretation of the data. Y. M. and X. T. drafted the paper. C. W. and L. H. supervised the study. All authors participated in revising the manuscript and approved the completed version. C. W. and L. H. are co-corresponding authors and contributed equally.

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Data availability
The data that support the findings of this study are derived from the Jinhua Healthcare Security Administration, which were used under authorization. Data are available from the authors upon reasonable request and with permission of the Jinhua Healthcare Security Administration.

Declarations

Ethics approval and consent to participate
The facilities gathered the data routinely in clinical practice according to the requirements of the local healthcare administration, and all data were processed without any personal identities being disclosed. Informed consent was obtained from the participants, and in the case of minors, which refers to individuals younger than the age of 16, consent for participation was obtained from their parents or legal guardians. Ethics approval of the study has been obtained from the Medical Ethics Committee of the Chinese Academy of Medical Sciences & Peking Union Medical College to report the data for research purposes (Protocol Number: X170315009). This study was conducted in accordance with the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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